MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY, ALLAHABAD
DATA STRUCTURE LAB
ASSIGNMENT- 8

1. You must represent two stacks in an array, memory[MEMORY SIZEJ, Write C functions that add (PUSH) and delete (POP) an item from stack i, 0<j<n. Your functions should be able to add elements to the stacks as long as the total number of elements in both stacks is less than MEMORY SIZE-1.

2. Given a singly linked list head with n nodes, write a function Rotate_K(head, k) that rotates every group of k nodes to the right in the linked list. The function should return the modified linked list. If the number of nodes is not a multiple of k, leave the remaining nodes as is. Rotation means shifting each node in the group one position to the right, with the last node moving to the front of the group.

Example 1:
Input: 123456789 and k =3
Output: 312-645-978

Example 2:
Input: 1-2-3-4-5-6-7-8 and k = 3
Output: 312-645-78